PROBABILITY THEORY
MTM2592-Gr:2

9TH WEEK

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SPECIAL PROBABILITY DISTRIBUTIONS

A probability function is denoted in a general form

$$\boxed{f(X; p_1, p_2, ..., p_k)}$$

where X is a random variable and $p_1, p_2, ..., p_k$ are called distribution parameters.

1. THE DISCRETE UNIFORM DISTRIBUTION

If a random variable can take on n different values with equal probability, we say that it has a **discrete uniform distribution**; symbolically, we have the following definition.

Definition: If the random variable X takes the values $x_1, x_2, \ldots, x_n$ with equal probabilities, then the discrete uniform distribution is given by

$$f(x; n) = \frac{1}{n}; \quad x = x_1, x_2, ..., x_n$$

Expected Value and Variance Of The Discete Uniform Distribution

$$E(X)=\mu=\sum_{x=1}^{n} x.f(x;n)=\sum_{x=1}^{n} x.\frac{1}{n}=\frac{\sum_{x=1}^{n} x}{n}$$

$$\sum_{x=1}^{n} x=1+2+3+....+n=\frac{n(n+1)}{2}$$

$$\boxed{E(X)=\frac{1}{n}.\frac{n(n+1)}{2}=\frac{n+1}{2}}$$

$$E(X^2)=\sum_{x=1}^{n} x^2.f(x;n)=\sum_{x=1}^{n} x^2.\frac{1}{n}=\frac{\sum_{x=1}^{n} x^2}{n}$$

$$\sum_{x=1}^{n} x^2=1^2+2^2+3^2+....+n^2=\frac{n(n+1)(2n+1)}{6}$$

$$E(X^2)=\frac{1}{n}.\frac{n(n+1)(2n+1)}{6}=\frac{(n+1)(2n+1)}{6}$$

$$\boxed{Var(X)=E(X^2)-[E(X)]^2=\frac{(n+1)(2n+1)}{6}-\left[\frac{n+1}{2}\right]^2=\frac{n^2-1}{12}}$$

Example:There are 10 identical balls numbered 1 to 10 in a bag. If X random variable shows the number of the balls,

a) What is the probability function of X?
b) E(X)=?, Var(X)=?

Solution: Since the probability of selecting balls is equal

$$\boxed{f(x;n) = \frac{1}{n}; \quad x = x_1, x_2, \ldots, x_n}$$

a) $$f(x;10) = \begin{cases} \frac{1}{10}; & x = 1, 2, \ldots\ldots, 10 \\ 0, & \text{elsewhere} \end{cases}$$

b) $$E(X) = \mu = \frac{n+1}{2} = \frac{10+1}{2} = 5,5$$

$$Var(X) = \frac{n^2-1}{12} = \frac{100-1}{12} = 8,25$$

2. THE BERNOULLI DISTRIBUTION

Let an experiment consist of repeated trials and each trial be independent. If each trial **has two possible outcomes**, i.e. Success or failure, each trial is called a Bernoulli trial. If the probability of an outcome is denoted by p, the Bernoulli distribution is given by

$$\boxed{f(x;p) = p^{x}(1-p)^{1-x} \qquad x = 0,1}$$

There is only one parameter p,

X: Number of successes in a Bernoulli experiment;

p+q=1

$$f(x;p) = \begin{cases} 1-p=q, & x=0 \\ p, & x=1 \\ 0, & \text{Other cases} \end{cases}$$

$$\text{i)}\quad f(x;p) = p^{x}(1-p)^{1-x} \geq 0$$

$$\text{ii)}\quad \sum_{x=0}^{1} f(x;p) = \sum_{x=0}^{1} p^{x}q^{1-x} = q+p = 1$$

Probability distribution

Note:

- There are only two possible results;
- The probability of occurrence of these results does not change from experiment to experiment

Moment Generating Function, Expected Value and Variance Of The Bernoulli Distribution

$$a)\ \ M(t) = E[e^{tX}] = \sum_{x=0}^{1} e^{tx} f(x;p) = \sum_{x=0}^{1} e^{tx} p^{x}(1-p)^{1-x}$$

$$= e^{t0} p^{0}(1-p)^{1-0} + e^{t1} p^{1}(1-p)^{1-1}$$

$$= (1-p) + e^{t}.p = q + e^{t}.p$$

If we differentiate M(t) first time and then set t=0, we obtain the first moment of X.

$$M'(t) = e^{t}.p$$

$$M'(0) = p$$

$$E(X) = \mu = p$$

The second moment is obtained by taking the second derivative of the moment generating function with respect to t, and then set t=0;

$$M''(t) = e^{t}.p$$

$$M''(0) = p$$

$$E(X^{2}) = p$$

So variance;

$$\mathrm{Var}(X) = M''(0) - [M'(0)]^2$$

$$= p - p^2 = p(1-p) = pq$$

b) By using the definition of the expected value and variance of an X random variable,

$$E(X) = \sum_{x=0}^{1} x.f(x;p) = \sum_{x=0}^{1} x.p^x(1-p)^{1-x}$$

$$= 0.p^0(1-p)^{1-0} + 1.p^1(1-p)^{1-1} = p$$

$$E(X^2) = \sum_{x=0}^{1} x^2.f(x;p) = \sum_{x=0}^{1} x^2.p^x(1-p)^{1-x}$$

$$= 0.p^0(1-p)^{1-0} + 1.p^1(1-p)^{1-1} = p$$

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = p - p^2 = p(1-p) = pq$$

Example:The probability of going to the cinema on Sunday night is 0.3 for Mehmet. Write the probability function of Mehmet's Sunday night.

Solution:

$$f(x;0.3)=\begin{cases}0,7, & x=0\\ 0,3, & x=1\\ 0, & \text{Elsewhere}\end{cases}$$

$$\boxed{f(x;p)=p^{x}(1-p)^{1-x} \quad x=0,1}$$

$$f(x;p)=\begin{cases}(0,3)^{x}(0,7)^{1-x} & x=0,1\\ 0, & \text{elsewhere}\end{cases}$$

Example:If the probability function of X random variable is given below.

$$f(x;p)=\begin{cases}\left(\frac{1}{4}\right)^{x}\left(\frac{3}{4}\right)^{1-x} & x=0,1\\ 0, & \text{elsewhere}\end{cases}$$

Find moment generating function of X, expected value of X and variance of X.

Solution: This is a Bernoulli distribution with parameter p=1/4

$$\boxed{f(x;p)=p^{x}(1-p)^{1-x} \qquad x=0,1}$$

$$M(t)=E[e^{tX}]=\sum_{x=0}^{1}e^{tx}\left(\frac{1}{4}\right)^{x}\left(\frac{3}{4}\right)^{1-x}=\frac{3}{4}+e^{t}\frac{1}{4}$$

$$E(X)=p=\frac{1}{4}$$

$$Var(X)=pq=\frac{1}{4}.\frac{3}{4}=\frac{3}{16}$$

3. BINOMIAL DISTRIBUTION (TWO TERM DISTRIBUTION)

Let **a Bernoulli trial be repeated n times** and the probability of an outcome be p. If X represents the number of successes of this outcome, then the binomial distribution is given by;

X: Number of successes in n Bernoulli experiments

$$\boxed{b(x;n,p) = \binom{n}{x} p^{x} (1-p)^{n-x} \qquad x = 0,1,2,\ldots\ldots,n}$$

n is an integer and $n \geq 1$

The binomial probability function has two parameters, such as **n** and **p**.

If n = 1, the Binom random variable becomes a Bernoulli random variable. The corresponding probabilities for X = 0,1,2,…., N,

$$\boxed{b(x;n,p)=\binom{n}{x}p^x(1-p)^{n-x}}$$

X	0	1	2	……….	n
b(x;n,p)	q^n	$\binom{n}{1}pq^{(n-1)}$	$\binom{n}{2}p^2q^{(n-2)}$		p^n

$$(p+q)^n=q^n+\binom{n}{1}pq^{(n-1)}+\binom{n}{2}p^2q^{(n-2)}+\ldots\ldots+p^n$$

The binomial random variable is 'the number of successes'.

$$\left.\boxed{\begin{array}{l} \text{i)}\ \ b(x;n,p)\geq 0 \\ \text{ii)}\ \sum_{x=0}^{n} b(x;n,p)=\sum_{x=0}^{n}\binom{n}{x}p^xq^{n-x}=(p+q)^n=1 \end{array}}\right\}\ \text{Probability distribution}$$

Moment Generating Function, Expected Value and Variance Of The Binomial Distribution

a) $M(t) = E[e^{tX}] = \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^x q^{n-x} = \sum_{x=0}^{n} \binom{n}{x} \left(e^t p\right)^x q^{n-x}$

$e^t.p=r$ taken. So

$$M(t) = \sum_{x=0}^{n} \binom{n}{x} r^x q^{n-x} = (r+q)^n = (pe^t+q)^n$$

Expected value and variance can be found by using moment generating function; E(X) is found by setting t = 0 in the first moment.

$$M'(t) = n(pe^t+q)^{n-1}.pe^t$$

$$M'(0) = E(X) = \mu = n\underbrace{(p+q)^{n-1}}_{1}.p = np$$

The second moment is obtained by taking the second derivative of the moment generating function with respect to t, and by setting t = 0.

$$M''(t) = n(n-1)(pe^t+q)^{n-2}.pe^t.pe^t + n(pe^t+q)^{n-1}.pe^t$$

$$= n(pe^t+q)^{n-2}.pe^t\left[(n-1)pe^t + (pe^t+q)\right]$$

$$= n(pe^t + q)^{n-2}.pe^t \left[(n-1)pe^t + (pe^t + q) \right]$$

$$M''(0) = n(p+q)^{n-2}.p\left[(n-1)p + (p+q)\right]$$

$$= np[(n-1)\,p+1]$$

$$= np(np - p + 1)$$

$$= np(np + q)$$

$$= n^2p^2 + npq$$

So variance,

$$Var(X) = M''(0) - [M'(0)]^2$$

$$= n^2p^2 + npq - n^2p^2 = npq$$

b) By using the definition of the expected value and variance of an X random variable,

$$E(X) = \mu = \sum_{x=0}^{n} x.b(x;n,p) = \sum_{x=0}^{n} x.\binom{n}{x} p^{x} q^{n-x}$$

$$= 0.\binom{n}{0} p^{0} q^{n-0} + \sum_{x=1}^{n} x.\binom{n}{x} p^{x} q^{n-x}$$

$$= \sum_{x=1}^{n} x.\binom{n}{x} p^{x} q^{n-x} = \sum_{x=1}^{n} x.\frac{n!}{(n-x)!x!} p^{x} q^{n-x}$$

$$= \sum_{x=1}^{n} \frac{n(n-1)!}{(n-x)!(x-1)!} p^{x} q^{n-x}$$

$$= np\sum_{x=1}^{n} \frac{(n-1)!}{(n-x)!(x-1)!} p^{x-1} q^{n-x}$$

By placing n-1=m and x-1=y

$$= np\sum_{y=0}^{m+1} \frac{m!}{(m-y)!y!} p^{y} q^{m-y}$$

$$= np[p+q]^{m}; \quad (p+q)^{m} = 1$$

E(X)=np

$$E(X^2) = \sum_{x=0}^{n} x^2 \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=1}^{n} x^2 . \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=1}^{n} x^2 . \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

$$= \sum_{x=1}^{n} npx \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} q^{n-x}$$

$$= np \sum_{x=1}^{n} (x-1+1) \frac{(n-1)!}{(n-x)!(x-1)!} p^{x-1} q^{n-x}$$

$$= np \left[\sum_{x=1}^{n} (x-1) \frac{(n-1)!}{(n-x)!(x-1)!} p^{x-1} q^{n-x} + \sum_{x=1}^{n} \frac{(n-1)!}{(n-x)!(x-1)!} p^{x-1} q^{n-x} \right]$$

The value of the second sum is 1.Because $(p+q)^{n-1}=1$.

$$= np \left[(n-1) p \sum_{x=2}^{n} \frac{(n-2)!}{(n-x)!(x-2)!} p^{x-2} q^{n-x} + 1 \right]$$

$$= np \left[(n-1) p \sum_{x=2}^{n} \binom{n-2}{x-2} p^{x-2} q^{n-x} + 1 \right]$$

$$\sum_{x=2}^{n}\binom{n-2}{x-2}p^{x-2}q^{n-x}=\sum_{x=0}^{n}\binom{n-2}{x}p^{x}q^{n-x-2}=\sum_{x=0}^{n-2}\binom{n-2}{x}p^{x}q^{n-x-2}=(p+q)^{n-2}=1$$

$$= np[(n-1)p+1]$$

$$= n^2p^2-np^2+np$$

$$= n^2p^2+npq$$

$$\mathrm{Var}(X)=E(X^2)-[E(X)]^2$$

$$= n^2p^2+npq-n^2p^2$$

$$= npq$$

c) Since the binomial distribution is the sum of **n independent** Bernoulli experiments; If Y is a Bernoulli variable, E [Y] = p. Using properties of the expected value properties,

$$
\begin{aligned}
E(X) = \mu = E\left[\sum_{i=1}^{n} Y_i\right] &= E[Y_1 + Y_2 + \dots + Y_n] \\
&= E[Y_1] + E[Y_2] + \dots + E[Y_n] \\
&= p + p + \dots\dots + p \\
&= np
\end{aligned}
$$

obtained. Using the variance properties with the same thought,

$$
\begin{aligned}
Var(X) = Var\left[\sum_{i=1}^{n} Y_i\right] &= Var[Y_1 + Y_2 + \dots + Y_n] \\
&= Var[Y_1] + Var[Y_2] + \dots + Var[Y_n] \\
&= pq + pq + \dots\dots + pq \\
&= npq
\end{aligned}
$$

Sometimes it may be necessary to deal with the success rate, not the number of successes in n trials. That is, if x indicates the number of successes, the distribution of x / n variables is in question. The arithmetic mean of this distribution;

$$E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}np = p$$

and variance

$$Var\left(\frac{X}{n}\right) = \frac{1}{n^2}Var(X) = \frac{1}{n^2}npq = \frac{pq}{n}$$

Theorem: $\boxed{b(x;n,p) = b(n-x;n,1-p)}$

Example A fair coin is tossed six times. What is the probability that exactly three heads turn up? The answer is

$$b(3,6,1/2)= \binom{6}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^3 = 20 \cdot \frac{1}{64} = .3125 .$$

Example A die is rolled four times. What is the probability that we obtain exactly one 6? We treat this as Bernoulli trials with *success* = "rolling a 6" and *failure* = "rolling some number other than a 6." Then $p = 1/6$, and the probability of exactly one success in four trials is

$$b(1,4,,1/6) = \binom{4}{1}\left(\frac{1}{6}\right)^1\left(\frac{5}{6}\right)^3 = .386 .$$

EXAMPLE

Find the probability of getting five heads and seven tails in 12 flips of a balanced coin.

Solution
Substituting $x = 5$, $n = 12$, and $p = \frac{1}{2}$ into the formula for the binomial distribution, we get

$$b\left(5; 12, \frac{1}{2}\right) = \binom{12}{5}\left(\frac{1}{2}\right)^5\left(1-\frac{1}{2}\right)^{12-5}$$

we find that the result is $792\left(\frac{1}{2}\right)^{12}$, or approximately 0.19.

EXAMPLE

Find the probability that 7 of 10 persons will recover from a tropical disease if we can assume independence and the probability is 0.80 that any one of them will recover from the disease.

Solution
Substituting $x = 7$, $n = 10$, and $p = 0.80$ into the formula for the binomial distribution, we get

$$b(7; 10, 0.80) = \binom{10}{7}(0.80)^7(1-0.80)^{10-7}$$

and we find that the result is $120(0.80)^7(0.20)^3$, or approximately 0.20.

Example: Let a balanced (fair) coin be tossed 5 times, find the probability that the coin comes up head

a) Two times,
b) At least 4 times,
c) If this experiment is repeated n times, what is the averagely expectation of the head numbers?

Solution: $$b(x;n,p)=\binom{n}{x}p^x(1-p)^{n-x} \qquad x=0,1,2,......,n$$

a) $$b\left(x;5,\frac{1}{2}\right)=\binom{5}{x}\left(\frac{1}{2}\right)^x\left(1-\frac{1}{2}\right)^{5-x}$$

$$P(X=2)=\binom{5}{2}\left(\frac{1}{2}\right)^2\left(1-\frac{1}{2}\right)^{5-2}=\frac{10}{32}\cong 0.3125$$

b) $$P(X\geq 4)=P(X=4)+P(X=5)$$

$$=\binom{5}{4}\left(\frac{1}{2}\right)^4\left(1-\frac{1}{2}\right)^{5-4}+\binom{5}{5}\left(\frac{1}{2}\right)^5\left(1-\frac{1}{2}\right)^{5-5}=\left(5+1\right)\frac{1}{2^5}=0.1875$$

c) $$E(X)=np=n.\frac{1}{2}=\left(0.5\right)n$$

Example: A doctor knows from her experiences that the probability of a patient suffering a specific disease will recover from this disease is 0.30. If 4 patients suffer from this disease, what is the probability that

a) All of them will recover,
b) At least 2 of them will recover?

Solution:

a) $$P(X=4)=b(4;4,0.30)=\binom{4}{4}(0.30)^4(0.70)^0=0.0081$$

b) $$P(X\geq 2)=P(X=2)+P(X=3)+P(X=4)$$

$$=\binom{4}{2}(0.30)^2(0.70)^2+\binom{4}{3}(0.30)^3(0.70)^1+\binom{4}{4}(0.30)^4(0.70)^0=0.3483$$

Example: A company is selected 120 products from a parcel of which 5% are defective. Let Y be the number of chosen defective products. If the repair cost of the defective products is given by

$$M = 8Y^2 + \frac{1}{2}Y - 10$$

Compute the expected repair cost.

Solution:

$$E(Y) = np = 120.(0,05) = 6$$

$$Var(Y) = npq = 120.(0,05).(0,95) = 5,7$$

$$E(M) = E\left(8Y^2 + \frac{1}{2}Y - 10\right) = 8E(Y^2) + \frac{1}{2}E(Y) - E(10)$$

$$Var(Y) = E(Y^2) - \left[E(Y)\right]^2 \Rightarrow 5,7 = E(Y^2) - 6^2 \Rightarrow E(Y^2) = 41,7$$

$$E(M) = 8.(41,7) + \frac{1}{2}.6 - 10 = 326,6$$

4. MULTINOMIAL DISTRIBUTION

Definition: There are n independent experiments with only two possible outcomes in the binomial distribution. Consider the fact that an experiment has more than two outcomes. If this experiment is repeated n times, then a Multinomial distribution is used. This distribution is also called Generalized Binomial Distribution.

x_1 represents the number of occurrence of A_1,
x_2 represents the number of occurrence of A_2,
.....
x_k represents the number of occurrence of A_k
then the probability $P(X_1 = x_1, X_2 = x_2, \ldots. X_k = x_k)$

$$\boxed{f(x_1, x_2, \ldots, x_k; n, p_1, p_2, \ldots p_k) = \binom{n}{x_1, x_2, \ldots, x_k} p_1^{x_1} p_2^{x_2} \ldots . p_k^{x_k}}$$

is called Multinomial distribution.

$$\binom{n}{x_1, x_2, \ldots, x_k} = \frac{n!}{x_1!.x_2!\ldots x_k!} \Rightarrow$$ Permutation with repetition

$$\text{For all i} \quad x_i \geq 0, \quad i = 1, \ldots, k \quad \sum_{i=1}^{k} x_i = n \text{ ve } \sum_{i=1}^{k} p_i = 1$$

$$E[X_i] = \mu_i = np_i \quad \text{and} \quad Var[X_i] = np_i q_i \quad i = 1, \ldots, k$$

For k = 2, the Multinomial distribution is reduced to the binomial distribution.

EXAMPLE

A certain city has 3 newspapers, A, B, and C. Newspaper A has 50 percent of the readers in that city. Newspaper B, has 30 percent of the readers, and newspaper C has the remaining 20 percent. Find the probability that, among 8 randomly-chosen readers in that city, 5 will read newspaper A, 2 will read newspaper B, and 1 will read newspaper C. (For the purpose of this example, assume that no one reads more than one newspaper.)

$$\boxed{f(x_1, x_2, \ldots, x_k; n, p_1, p_2, \ldots p_k) = \binom{n}{x_1, x_2, \ldots, x_k} p_1^{x_1} p_2^{x_2} \ldots p_k^{x_k}}$$

Solution

Substituting $x_1 = 5$, $x_2 = 2$, $x_3 = 1$, $p_1 = 0.50$, $p_2 = 0.30$, $p_3 = 0.20$, and $n = 8$ into the formula of multinomial distribution, we get

$$f(5, 2, 1; 8, 0.50, 0.30, 0.20) = \frac{8!}{5! \cdot 2! \cdot 1!}(0.50)^5(0.30)^2(0.20)$$
$$= 0.0945$$

Example: There are 5 black, 4 red and 2 green balls in a box. If 6 balls are selected with replacing them back, find the probability that selected balls consists of 2 black, 3 red and 1 green balls?

Solution:

$$P(X_1 = 2, X_2 = 3, X_3 = 1) = ?$$

$$n = 6$$

$$p_1 = \frac{5}{11}, \quad p_2 = \frac{4}{11}, \quad p_3 = \frac{2}{11}$$

$$x_1 + x_2 + x_3 = 6$$

$$\boxed{f(x_1, x_2, \ldots., x_k; n, p_1, p_2, \ldots p_k) = \binom{n}{x_1, x_2, \ldots, x_k} p_1^{x_1} p_2^{x_2} \ldots. p_k^{x_k}}$$

$$f\left(2,3,1;6,\frac{5}{11},\frac{4}{11},\frac{2}{11}\right) = \binom{6}{2,3,1}\left(\frac{5}{11}\right)^2\left(\frac{4}{11}\right)^3\left(\frac{2}{11}\right) = \frac{6!}{2!3!1!}\left(\frac{5}{11}\right)^2\left(\frac{4}{11}\right)^3\left(\frac{2}{11}\right)$$

5. GEOMETRIC DISTRIBUTION

Definition: Consider a Bernoulli trial. If X represents the number of trials on which the **first success occurs**, then the geometric distribution is given by

$$\boxed{g(x;p) = q^{x-1}p \qquad x = 1, 2, 3, \ldots}$$

X: Number of Bernoulli trials required to achieve a success

The parameter of this probability distribution is **p**. Let's show that g (x; p) is a probability function:

i) $g(x;p) \geq 0$

ii) Since this is a geometric sequence,

$$\sum_{x=1}^{\infty} g(x;p) = \sum_{x=1}^{\infty} q^{x-1}p = p(1+q+q^2+q^3+\ldots.) = p\frac{1}{1-q} = 1$$

Probability distribution

Moment Generating Function, Expected Value and Variance Of The Geometric Distribution

a) By using the definition of moment generating function;

$$M(t) = E[e^{tX}] = \sum_{x=1}^{\infty} e^{tx} pq^{x-1} = p\sum_{x=1}^{\infty} e^{tx} q^{x-1} = \frac{p}{q}\sum_{x=1}^{\infty} e^{tx} q^{x}$$

$$= \frac{p}{q}\sum_{x=1}^{\infty}\left(e^{t}q\right)^{x} = \frac{p}{q}\left[e^{t}q + \left(e^{t}q\right)^{2} + \ldots\ldots\right]$$

Sum of the geometric sequence in square brackets

$$e^{t}q.\frac{1}{1-e^{t}q}$$

$$M(t) = \frac{p}{q}e^{t}q\frac{1}{1-e^{t}q} = \frac{pe^{t}}{1-e^{t}q}$$

$$M'(t) = \frac{pe^{t}(1-e^{t}q) + pqe^{2t}}{\left(1-e^{t}q\right)^{2}}$$

$$M'(0) = E(X) = \frac{p(1-q)+pq}{\left(1-q\right)^{2}} = \frac{p^{2}+pq}{p^{2}} = \frac{p\left(p+q\right)}{p^{2}} = \frac{1}{p}$$

$$M''(0) = E(X^2) = \frac{2q + p}{p^2}$$

So variance,

$$\mathrm{Var}(X) = M''(0) - [M'(0)]^2$$

$$\mathrm{Var}(X) = \frac{2q + p}{p^2} - \frac{1}{p^2} = \frac{q}{p^2}$$

b) By using the definition of the expected value and variance of an X random variable,

$$E(X)=\mu=\sum_{x=1}^{n}x.p(x)=\sum_{x=1}^{n}x.q^{x-1}.p=p.\frac{d}{dq}\left[\sum_{x=1}^{\infty}q^{x}\right]$$

$$=p.\frac{d}{dq}\left[\frac{q}{1-q}\right]=p\left[\frac{1.(1-q)-q(-1)}{(1-q)^{2}}\right]=p\frac{1}{(1-q)^{2}}=\frac{1}{p}$$

$$E(X^{2})=\sum_{x=1}^{n}x^{2}.p\,q^{x-1}=p\sum_{x=1}^{n}\left(x\left(x-1\right)+x\right)q^{x-1}=pq\sum_{x=1}^{n}\left(x\left(x-1\right)+x\right)q^{x-2}=pq\sum_{x=1}^{n}\left(x\left(x-1\right)\right)q^{x-2}+E(X)$$

$$pq\frac{d^{2}}{dq^{2}}\left[\sum_{x=1}^{\infty}q^{x}\right]+\frac{1}{p}=pq\frac{d^{2}}{dq^{2}}\left[\frac{q}{1-q}\right]+\frac{1}{p}=pq\frac{d}{dq}\left[\frac{1}{(1-q)^{2}}\right]+\frac{1}{p}=pq\left[\frac{2}{(1-q)^{3}}\right]+\frac{1}{p}$$

$$=pq\left[\frac{2}{(1-q)^{3}}\right]+\frac{1}{p}=pq\left[\frac{2}{p^{3}}\right]+\frac{1}{p}=\frac{2q}{p^{2}}+\frac{1}{p}=\frac{2q+p}{p^{2}}$$

$$Var(X)=\frac{2q+p}{p^{2}}-\frac{1}{p^{2}}=\frac{2q}{p^{2}}+\frac{p}{p^{2}}-\frac{1}{p^{2}}=\frac{2q}{p^{2}}-\frac{q}{p^{2}}=\frac{q}{p^{2}}=\frac{1-p}{p^{2}}$$

EXAMPLE

If the probability is 0.75 that an applicant for a driver's license will pass the road test on any given try, what is the probability that an applicant will finally pass the test on the fourth try?

Solution

Substituting $x = 4$ and $p = 0.75$ into the formula for the geometric distribution, we get

$$\begin{aligned} g(4; 0.75) &= 0.75(1 - 0.75)^{4-1} \\ &= 0.75(0.25)^3 \\ &= 0.0117 \end{aligned}$$

Of course, this result is based on the assumption that the trials are all independent, Example: Let a die be rolled until it comes up 1.

a) For the number of trials, define an appropriate probability function,

b) Find the probability that the die will come up 1 at the third trial.

Solution:

a) $$f(x) = g\left(x; \frac{1}{6}\right) = \left(\frac{5}{6}\right)^{x-1} \frac{1}{6} \qquad x = 1, 2, 3, \ldots$$

b) $$P(X = 3 = g\left(3; \frac{1}{6}\right) = \left(\frac{5}{6}\right)^{3-1} \frac{1}{6} = \frac{25}{216}$$

6. NEGATIVE BINOMIAL (PASCAL) DISTRIBUTION

Definition: Consider a Bernoulli trial. If X represents the number of trials on which the **k-th success occurs**, then the negative binomial distribution is given by

$$\boxed{b^*(x;k,p)=\binom{x-1}{k-1}p^k q^{x-k} \qquad x=k,k+1,k+2,\ldots}$$

which means P(X=x). It is also called Pascal probability function.

X: Number of Bernoulli trials required to achieve k success

while k-1 successful result will be obtained from x-1 experiment, k-th success will be achieved in experiment x. Negative Binomial distribution is the general form of Geometric distribution.

Let's investigate whether b * is a probability function.

$$\boxed{\binom{n}{k}=\binom{n}{n-k}}$$

$$\text{i) For } \forall x \quad b^*(x;k,p)\geq 0 \qquad \boxed{x-k=t}$$

n is **not a natural number**

$$\text{ii) } \sum_{x=k}^{\infty}\binom{x-1}{k-1}p^k q^{x-k}=\sum_{t=0}^{\infty}\binom{t+k-1}{k-1}p^k q^t=\sum_{t=0}^{\infty}\binom{t+k-1}{t}p^k q^t=$$

$$\boxed{(a+b)^n=\sum_{r=0}^{\infty}\binom{n}{r}a^{n-r}b^r}$$

$$=p^k(1-q)^{-k}=1$$

$$\boxed{\binom{-r}{k}=(-1)^k\binom{r+k-1}{k}}$$

$$\boxed{\sum_{t=0}^{\infty}\binom{t+k-1}{t}(q)^t=\sum_{t=0}^{\infty}\binom{-k}{t}(-q)^t=(1-q)^{-k}}$$

Note: Negative binomial distribution with k = 1 turns into a geometric distribution.

Expected Value and Variance Of The Negative Binomial Distribution

By using the definition of the expected value and variance of an X random variable,

X_1: Number of attempts required until first success
X_2: The number of attempts required until the second success after the first success
.
X_k: The number of attempts required until the k^{th} success after the k-1 success

X: The number of experiments required to achieve k successful results
$X = X_1 + X_2 + \dots\dots + X_k$

X_i are the random variables of geometric distribution

$$E[X_i] = \frac{1}{p}$$

$$E[X] = E[X_1] + E[X_2] + \dots . + E[X_k]$$

$$= \frac{1}{p} + \frac{1}{p} + \dots\dots + \frac{1}{p} = k . \frac{1}{p}$$

For variance

$$Var[X_i] = \frac{q}{p^2}$$

$$Var[X] = Var[X_1] + Var[X_2] + ... + Var[X_k]$$

$$Var[X] = \frac{kq}{p^2}$$

Note:

$$\boxed{b^*(x;k,p) = \frac{k}{x}.b(x;k,p)}$$

EXAMPLE

If the probability is 0.40 that a child exposed to a certain contagious disease will catch it, what is the probability that the tenth child exposed to the disease will be the third to catch it?

Solution

Substituting $x = 10$, $k = 3$, and $p = 0.40$ into the formula for the negative binomial distribution, we get

$$\boxed{\begin{aligned} &b^*(x;k,p) = \binom{x-1}{k-1} p^k q^{x-k} \\ &x = k, k+1, k+2, \ldots \end{aligned}}$$

$$b^*(10; 3, 0.40) = \binom{9}{2}(0.40)^3(0.60)^7$$
$$= 0.0645$$

Use Theorem $\boxed{b^*(x;k,p) = \frac{k}{x}.b(x;k,p)}$ to rework Example

Solution

Substituting $x = 10$, $k = 3$, and $p = 0.40$ into the formula

$$b^*(10; 3, 0.40) = \frac{3}{10} \cdot b(3; 10, 0.40)$$
$$= \frac{3}{10}(0.2150)$$
$$= 0.0645$$

$$b(x;n,p) = \binom{n}{x} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \ldots\ldots, n$$

Example: 10% of the goods produced in a factory are defective. In a quality control process, find the probability that

a) The second defective good is the seventh good that have been checked,
b) More than 7 goods to be checked in order to obtain the second defective good.

Solution:
$$\boxed{b^*(x;k,p) = \binom{x-1}{k-1} p^k q^{x-k} \qquad x = k, k+1, k+2, \ldots}$$

a) $$P(X=2) = b^*(7;2,0.1) = \binom{7-1}{2-1}(0.1)^2(0.9)^{7-2} = 0.0354 \qquad x = k, k+1, k+2, \ldots$$

b) $$P(X>7) = 1 - P(X \le 7) = 1 - \left[P(X=2) + P(X=3) + \ldots\ldots + P(X=7)\right] = 0.8503$$